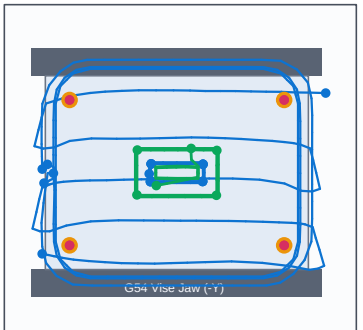
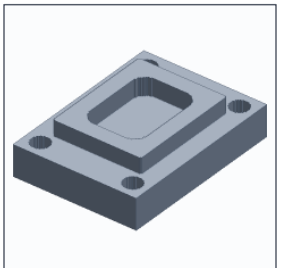
(a) Automated .stl -> 4-Tool TAMP Plan (O002001)

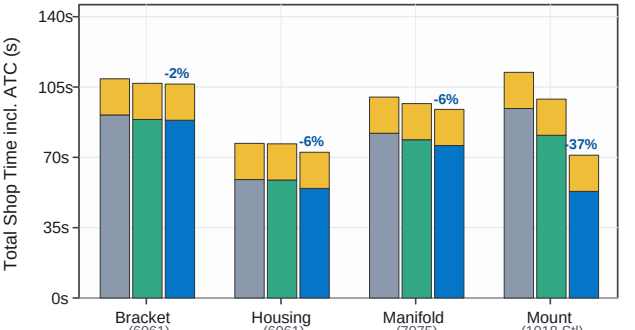




3D Carved Stock (4 Tools)

0.00 mm3 Gouge | Safety PASS

(b) Multi-Tool TAMP Total Cycle Time Across Alloys (Exp6)



T1: 10mm Rough/Face | **T2: 6mm Rest-Finish**

T3: 90deg Spot | **T4: 4mm G83 Peck**

Naive CAM | **Same-Speed**

Opt-Chipload | **ATC Swap (6s/chg)**